

Current Score : 33 / 37 Due : Monday, August 13, 2018 11:00 AM SAST Last Saved : n/a Saving... ()

The due date for this assignment is past. Your work can be viewed below, but no changes can be made.

Important! Before you view the answer key, decide whether or not you plan to request an extension. Your Instructor may *not* grant you an extension if you have viewed the answer key. Automatic extensions are not granted if you have viewed the answer key.

[View Key](#)

1. 3/3 points | [Previous Answers](#)SCalcET8M 5.3.016.

Use Part 1 of the [Fundamental Theorem of Calculus](#) to find the derivative of the function.

$$y = \int_1^{\tan(x)} \sqrt{5t + \sqrt{t}} \, dt$$

$y' =$

$\sec^2(x)\sqrt{5\tan(x)+\sqrt{\tan(x)}}$



Need Help?

Read It

Watch It

2. 3/3 points | [Previous Answers](#)SCalcET8M 5.3.018.

Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of the function.

$$y = \int_{4-3x}^3 \frac{u^3}{1+u^2} du$$

$y' =$

3(4-3x)³ / (1+(4-3x)²)



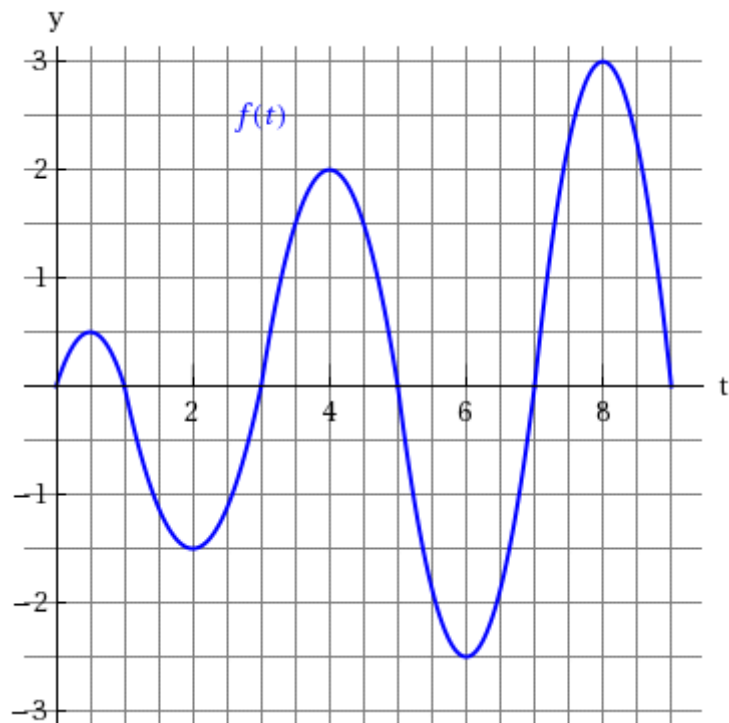
Need Help?

Read It

Watch It

3. 12/13 points | [Previous Answers](#)SCalcET8M 5.3.073.

Let $g(x) = \int_0^x f(t) dt$, where f is the function whose graph is shown.



(a) At what values of x do the local maximum and minimum values of g occur?

$x_{\min} =$ ✓ (smaller x -value)

$x_{\min} =$ ✓ (larger x -value)

$x_{\max} =$ ✓ (smaller x -value)

$x_{\max} =$ ✓ (larger x -value)

(b) Where does g attain its absolute maximum value?

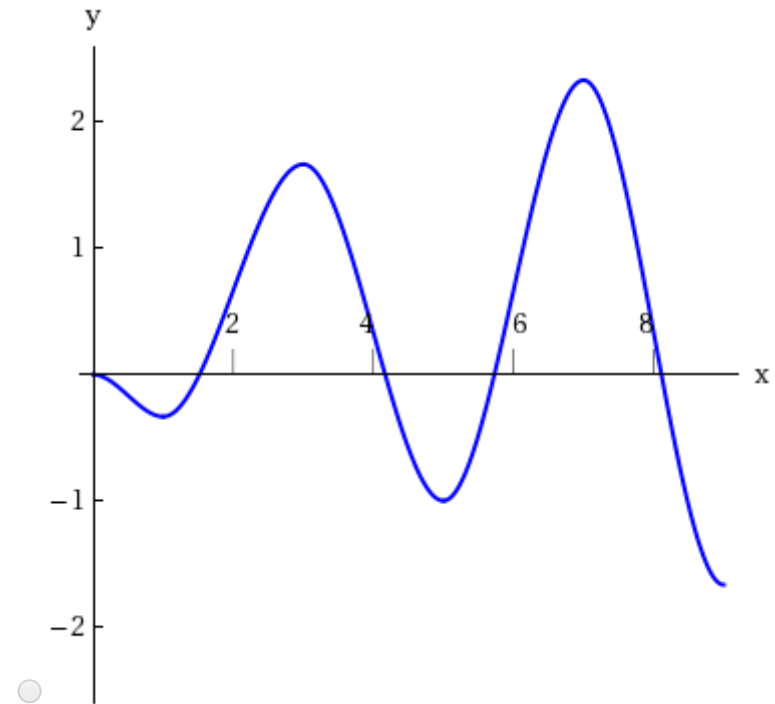
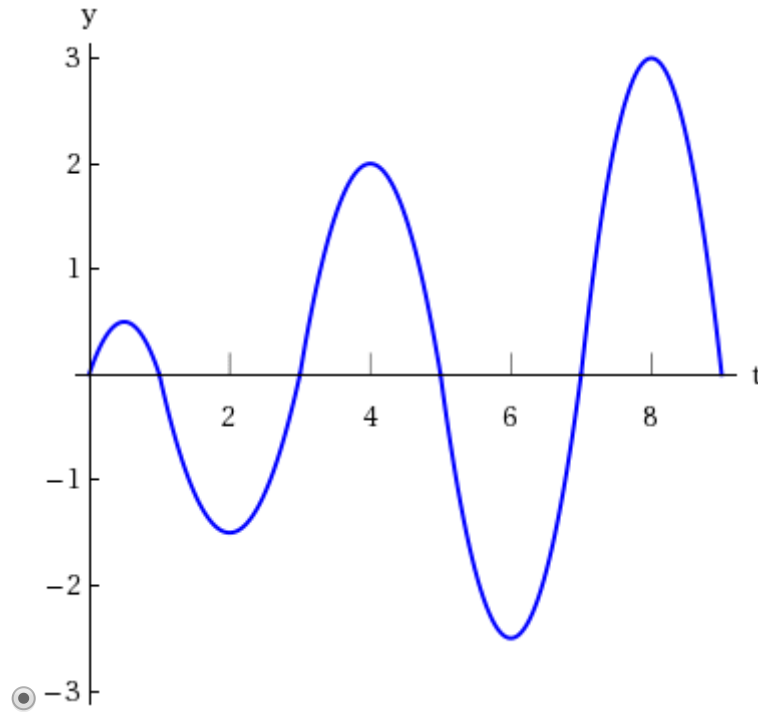
$x =$ ✓

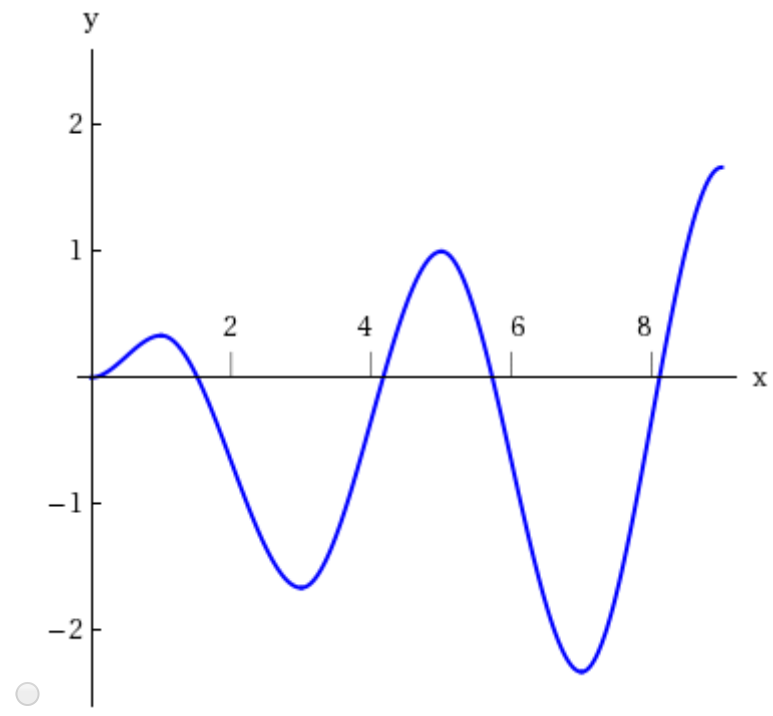
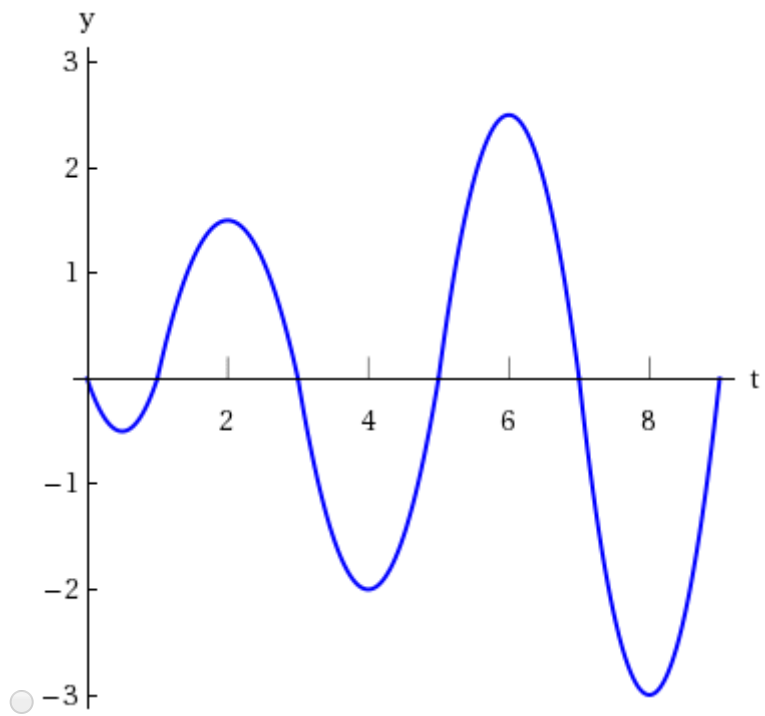
(c) On what interval is g concave downward? (Enter your answer using interval notation.)

$[(0.5, 2) \cup (4, 6) \cup (8, 9)]$



(d) Sketch the graph of g .





Enhanced Feedback

Please try again. Remember, $g(x)$ is the area under the curve of f between 0 and x . The local maximum values of g occur whenever the area under f stops increasing and starts decreasing. Similarly, the local minimum values of g occur whenever the area under f stops decreasing and starts increasing.

To find the intervals on which g is concave downward, look for the intervals on which $g''(x)$ is negative. Remember, $g''(x) = f'(x)$.

Need Help?

Read It

Watch It

4. 3/3 points | [Previous Answers](#)SCalcET8M 5.4.038.MI.

Evaluate the integral.

$$\int_0^{2\pi/3} \frac{3 \sin(\theta) + 3 \sin(\theta) \tan^2(\theta)}{\sec^2(\theta)} d\theta$$

\$\$92



Need Help?

Read It

Master It

5. 3/3 points | [Previous Answers](#)SCalcET8M 5.4.043.MI.

Evaluate the integral.

$$\int_0^{1/\sqrt{3}} 4 \frac{t^2 - 1}{t^4 - 1} dt$$

\$\$2\pi 3



Need Help?

Read It

Watch It

Master It

6. 3/3 points | [Previous Answers](#)SCalcET8M 5.4.046.

Evaluate the integral.

$$\int_0^{3\pi/2} 2|\sin(x)| \, dx$$

\$\$6



Need Help?

Read It

7. 3/6 points | [Previous Answers](#)SCalcET8M 5.4.059.

The velocity function (in meters per second) is given for a particle moving along a line.

$$v(t) = 3t - 8, \quad 0 \leq t \leq 3$$

(a) Find the displacement.

 m

(b) Find the distance traveled by the particle during the given time interval.

 m

Enhanced Feedback

Please try again. Because the particle changes direction during the time interval, the displacement will not be equal to the distance traveled. The displacement is simply the integral of the velocity function. The function for total distance traveled, however, treats any motion as positive, so it is the integral of the velocity function's absolute value.

Need Help?

Read It

Watch It

8. 3/3 points | [Previous Answers](#)SCalcET8M 5.4.064.

Water flows from the bottom of a storage tank at a rate of $r(t) = 200 - 4t$ liters per minute, where $0 \leq t \leq 50$. Find the amount of water that flows from the tank during the first 30 minutes.

 liters

Need Help?

Read It